Outlier Detection in Sensed Data using Statistical Learning Models for IoT

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***Abstract*—Internet of Things (IoT) devices are composed of millions of sensors that continuously sense environmental parameters which are effectively fused to gain insights on a particular area or region based on which desired actions are taken. However, before any fusion of the data takes place, checking the quality of data is of paramount importance as these data may also be contaminated with outliers which limit their overall efficiency. Outliers or in other words, any abrupt change in values can be either due to a drastic change in environment (Event) or because of any malfunctioning of the sensor (Error). This paper introduces an IoT architecture to detect the occurrence of both Error and Event in a forest environment with the help of four statistical models, i.e., Classification and Regression Trees (CART), Random Forest (RF), Gradient Boosting Machine (GBM) and Linear Discriminant Analysis (LDA). We take into account the spatial and temporal dependencies of the data as well. Simulation results show that the models can effectively detect both types of outliers with upto 100% accuracy for Error detection and upto 98.51% in case of Event detection. Also, the results provide evidence that using spatial and temporal parameters in event detection can have a positive influence on the prediction accuracy which was observed to increase from 81.68% to 96.53% using RF. Among the four models, RF is seen to outperform the others.**

***Index Terms*—Outlier detection, sensor, statistical, classification, IoT**

I. Introduction

The Internet of Things (IoT) is the upcoming trend in technology that will make our lives much more comfortable where everyday objects from personal hand-held equipments to industrial machinery are capable of generating data with the help of built-in sensors and taking decisions on the basis of that data. The vision of large scale IoT is to allow devices to sense and transmit their information among "things" in any monitoring environment. Needless to mention, with it comes the massive amount of data. This means new data management and integration approaches are to be adopted as any such scheme won't be of use if the data isn't accurate, secure, and actionable. Therefore, appropriate measures need to be taken to ensure that the data generated from the sensors are reliable before any decision is taken. Unreliable data can be an outcome of sensor fault (Error) or a rare observation (Event). Traditional outlier identification schemes fail to differentiate between Errors or Events and consider all outlier to be Errors, thus losing vital hidden information regarding rare events in the process. Hence, in this paper we have clearly distinguished both and conducted experiments on them separately. Since it is true that faulty sensors are also capable of generating similar event-type outliers. Thus, it becomes difficult to distinguish between the two types of outliers only by examining data from one node only. Hence, effective outlier detection techniques should incorporate spatial similarity between neighbouring nodes in order to detect events to avoid cases where wrong data generated is indeed due to a faulty sensor and not on account of an event. It is observed that errors caused due to sensor faults are considered to be spatially unrelated whereas events caused due to actual phenomenon are more likely to be spatially correlated [1]. In this work, four statistical models, i.e., Classification and Regression Trees (CART), Random Forest (RF), Gradient Boosting Machine (GBM) and Linear Discriminant Analysis (LDA) are used for detecting both type of outliers. The models are first trained with the training data after which the process of detection is carried out.
The rest of this paper is organized as follows. Section II discusses some of the previous works in this field. Section III introduces the proposed architecture for detecting outliers in the context of IoT with its preliminary definitions. Experiment and Analysis are presented in Section IV and finally Section V concludes our work with some remarks.

II. Outlier detection previous work

Identification and correction of outliers in the context of sensor data has long been an area of research for statisticians. The authors in [2], [3] have provided extensive surveys of outlier detection, specifying the different methodologies adopted for the purpose such as statistical approaches, clustering techniques, nearest neighbour and classification to name a few. Classification model works on the concept of first learning the model using a training set which includes both the normal data as well as outlier data during the training phase. After which, during the testing phase, it classifies previously unknown test data based on the learning model. The practice of training a model is known as learning and can be grouped into two categories, i.e., supervised and unsupervised learning. In supervised learning, we have prior knowledge about the distinctive behaviours of the classes i.e. both input data and output data are known. On the other hand, unsupervised learning method relies only upon the

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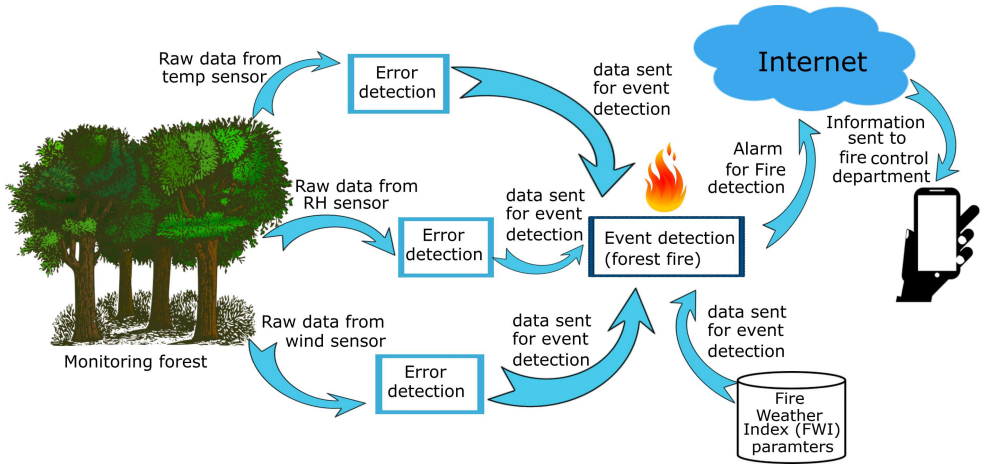


Fig. 1: IoT system architecture for forest fire detection

knowledge of input data and not on the output data where an attempt to classify input data is made during the testing phase that best describes the state. Our models in this paper adopt supervised learning methodology for classification using statistical techniques. Statistical classification makes use of the sensor data so that the best discrimination for classification is chosen for analysis based upon the sensor's response to past data. Works implementing classification model include [4]–[6] which adopts Bayesian networks (BN) and [7]–[9] use neural networks for classification as well as discriminant analysis, tree-based modelling and nearest-neighbour classification. Other works in this context are algorithms that mostly exploit the correlation in sensory data that may either be temporal [10] or spatial [11]. Also in [12] the authors exploited fuzzy theory for clustering in order to detect outliers which takes both spatial and temporal characteristics of sensor data. Next, a mechanism based on distance similarity is presented in [13] that reduces communication overhead among neighbouring nodes. This technique does not provide the flexibility to adapt itself to any network structure. The work in [14] also is based on distance-based detection of global outliers that works well with continuous query-based processing in sensor networks. In contrast with the past works, this paper presents a novel approach of integrating IoT in the context of outlier detection by discriminating between Errors and Events. Moreover, this work is focussed on a forest fire application where this distinction of Error and Event has never been attempted before in the past literature.

III. Proposed Architecture for Sensor Outlier Detection in IoT

An outlier or anomaly is defined as any observation that appears to differ significantly from other observations in a sample [2]. However, the mere existence of outliers is not enough proof that the data is corrupt. Outliers are, in fact, known to convey important information. Based on the underlying knowledge, an outlier could represent error or event which are defined next.

Definition 1. *Error denotes any noisy data that may have originated from a faulty sensor. They are characterized by any abrupt change in the measurements that differ significantly from the rest of the sample data.*

Definition 2. *Event refers to any phenomenon that changes the consistent state of the real-world. Outliers caused by events are found to occur less frequently than that caused by errors [16]. However, the duration of this type of outlier once detected is longer.*

Since our proposed architecture distinguishes between Error and Event outliers, the statistical models are run twice, once for Error detection, and once for Event detection. The four classification models that we used for the purpose of comparison in this work are CART, RF, GBM and LDA. Classification and regression trees is a popular machine learning technique used for the purpose of constructing classification models from data. The construction of the model is carried out by first partitioning the data space iteratively and then building a prediction model within each partition. The resulting partitioning can be viewed in the form of a decision tree. The creation of the classification trees also takes into account the prediction error which is the misclassification cost. In Random Forest, multiple trees are constructed as opposed to a single tree in CART model [15]. To classify an unknown sample based on its distinguishing features or attributes, each tree is expected to vote for a particular classification that it deems fit for the unknown sample. The classification which has the most number of votes in the forest is chosen as the final class. Gradient boosting or simply, GBM, is another machine learning model for regression and classification problems. It operates by constructing a prediction model in the form of an ensemble of weak prediction models, typically decision trees. The model is built in a stage-wise manner like other boosting methods. The model is finally generalized by

TABLE I: Dataset description

Feature Selection	Attribute	Description	Values
Spatial parameters (S)	X	x-axis coordinate in a $9 * 9$ grid	1 to 9
	Y	y-axis coordinate in a $9 * 9$ grid	1 to 9
Temporal parameters (T)	month	Month of the year	Jan to Dec
	day	weekday/weekend	0/1
Fire Weather Index (FWI)	FFMC	Fine Fuel Moisture Code	18.7 to 96.20
	DMC	Duff Moisture Code	1.1 to 291.3
	DC	Drought Code	7.9 to 860.6
	ISI	Initial Spread Index	0.0 to 56.10
Meteorological parameters (M)	temp	Outside temperature (in C)	2.2 to 33.30
	RH	Outside relative humidity (in %)	15 to 100
	wind	Outside wind speed (in km/h)	0.40 to 9.40
	rain	Outside rain (in mm/m2)	0.0 to 6.4
Response	area	Fire detected/ not detected	1/0

optimizing an arbitrary differentiable loss function. Linear discriminant analysis (LDA), which is generalization of the popular Fishers linear discriminant, works by finding a linear combination of features that distinguishes two or more classes of features or parameters. The resulting combination may be used as a linear classifier. For more information regarding the models, the readers are directed to refer to [15].

The proposed architecture for outlier detection in forest environment using IoT is presented in Fig. 1. As can be seen, the raw data from meteorological sensors are first validated against any Error due to sensor faults. This test is only performed for data prone to sensor faults. Other parameters which are not generated from sensors need not undergo the Error detection test. After the Error detection phase if an Error is detected, Event detection test need to be carried out to absolutely ascertain the reason of the Error. Since the cause of Error can be either due to a malfunctioning sensor or an alarming event, the Event detection test result is the deciding factor regarding the status of outlier in the monitoring environment. If the Event detection test result is true, then this ensures that the outlier data in the sensor reading in the form of previously conceived error, is not due to a malfunctioning sensor, but is an outcome of an Event. Thus, in such cases, necessary actions to mitigate the effect is taken which involves alerting the fire department and the concerning authorities monitoring the forest. If, however, the test of Event detection test is false, then it is an indication of the fact that the sensor is faulty and suitable measures have to be taken to immediately disconnect it with the network or to replace the erroneous sensor at the earliest. In this work, for the purpose of Event detection, we have performed some feature selection where similar features are grouped together and labelled accordingly. The X and Y coordinates together forms the spatial parameters (abbreviated as S), the temporal parameters (abbreviated as T) comprises of the month and day attributes, the weather parameters Fine Fuel Moisture Code (FFMC), Duff Moisture Code (DMC), drought code (DC) and Initial spread index (ISI) are collectively called as the Fire Weather Index (FWI) feature selection. Lastly, the temperature, relative humidity, wind and rain forms the meteorological parameters (abbreviated as M)(Refer Table I). The experiments are run for features FWI, STFWI, STM and M. For Event detection, the original dataset was divided such that the training set consists of 80% of the samples and the testing set contains 20% of the samples. The models were first trained with the training set after which the remaining 20% of the dataset was used for testing. For Error detection, the original dataset was infused with artificially created outliers after which it was similarly divided into training and test set in the ratio of 80:20. It is to be noted that Event detection tests are carried out only if an Error is detected, otherwise the models need not run the Event detection tests. This is based on the assumption that it is very unlikely for all sensors to malfunction simultaneously. In such a worst case scenario, the Event detection tests will detect an Event because of all malfunctioning sensors, and thus will generate a false positive result. However, in real life applications with thousands of sensors scattered in an environment, this situation has very low probability of occurrence and thus can be ignored in the present scenario.

IV. Experiment and Analysis

A. Experiment

In this work, we consider forest fire data [17] for the purpose of detecting outliers. The data was collected from a park in the North-east region of Portugal and consists of 517 entries from January to December from 2000 to 2003. All information regarding the dataset is tabulated in Table I. For validating our results, we have focused on

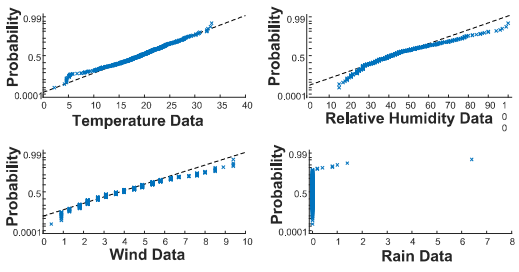


Fig. 2: Probability Plot for Normal distribution

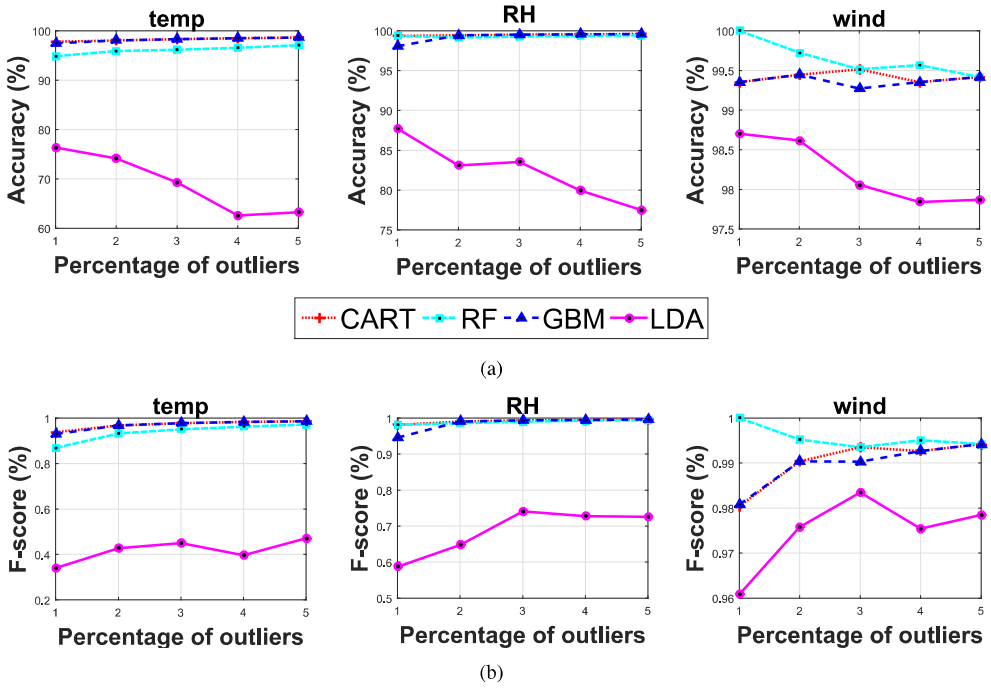


Fig. 3: Error detection for different % of corrupted data (outliers) in terms of (a) Accuracy (b) F-score

a fire detection application using four statistical learning models for each case of error (sensor fault) and event (important phenomenon) detection. They are Classification and Regression Trees (CART), Random Forest(RF), Gradient Boosting Machine (GBM) and Linear Discriminant Analysis (LDA) [15]. In order to conform to our classification models, the response attribute in the dataset area underwent some preprocessing where the values not equal to 0 were converted to 1 to depict the occurrence of fire. Thus the two classes are {fire(1), no fire(0)}. In addition, the value of day attribute were changed to 0 and 1 to represent weekday and weekend respectively. All simulations were conducted in MATLAB using statistical toolboxes. For RF model, training was done setting the number of trees to 350, after which no significant increase in the prediction accuracy was observed. The tuning parameters for GBM model were set as: shrinkage Factor = 0.1, sub-sampling Factor = 0.5, maxTreeDepth = 10, No of iterations = 350. The algorithm effectively chooses the best combinations of the parameters as well as the number of trees in the forest from the training samples. LDA model works on the assumption that the underlying distribution follows normality behaviour. To validate our assumption, a normality plot was generated for the four meteorological parameters i.e., temp, RH, wind and rain which is illustrated in Fig 2. By visually inspecting the graph we can conclude that the assumption of normality is quite reasonable. Plots for the four attributes are observed to fall close to the straight line indicating normal behaviour. After training the statistical models, they were evaluated using the well-known confusion matrix where the values of True Positive (TP), False Positive (FP), True Negative (TN) and False Negative (FN) were used to calculate the following other performance metrics :
$Accuracy = (TP + TN)/(TP + TN + FP + FN)$,
$Recall = TP/(TP + FN)$,
$False - positive - rate(FPR) = FP/(TN + FP)$,
$Specificity = TN/(TN + FP)$,
$Precision = TP/(FP + TP)$ and
$F-score = (2*Precision*Recall)/(Precision+Recall)$.

B. Analysis of Error detection

In our experiment, as ground truth for regular sensor nodes for the dataset was missing, we injected artificially-created outliers and tested the resulting dataset with our models to detect those outliers. Varying percentage of outliers (10%, 20%, 30%, 40%, 50% of the training set) were infused and tests were carried out for each degree of corrupted samples. This was done only for those attributes that were collected from sensors i.e. temp, RH and wind. After the testing procedure, promising results were witnessed with accuracies upto 100% for wind parameter tested with RF model with 10% outliers. Although CART, RF and GBM performs exceedingly well (above 94%), LDA performs poor in comparison. The best detection accuracy for LDA was 98.70% for wind, but with other weather parameters (temp, RH) the highest observed accuracy was only 87.70%. Fig 3(a) and 3(b) illustrates the performance of the four models in terms of Accuracy and F-score respectively where it is evident that even though CART, RF and GBM seem to result in somewhat consistent accuracies for all instance of the test samples, but GBM outperforms them all. It

TABLE II: Performance Evaluation for Event detection using various combinations of feature selection

	Models	Accuracy (%)	Recall (%)	FPR(%)	Precision(%)	Specificity(%)
FWI	CART	78.71	80.00	22.55	77.67	79.80
	RF	81.68	81.00	**17.65**	**81.82**	81.55
	GBM	**82.18**	**86.00**	21.57	79.63	**85.11**
	LDA	57.43	82.00	66.67	54.67	65.38
STFWI	CART	85.64	87.00	15.69	84.47	86.87
	RF	**96.53**	**100.00**	**6.86**	**93.46**	**100.00**
	GBM	89.11	96.00	17.65	84.21	95.45
	LDA	56.44	70.00	56.86	54.69	59.46
STM	CART	84.65	85.00	15.69	84.16	85.15
	RF	**98.51**	**100.00**	**2.94**	**97.09**	**100.00**
	GBM	88.12	91.00	14.71	85.85	90.63
	LDA	58.91	72.00	53.92	56.69	62.67
M	CART	87.62	87.00	11.76	87.88	87.38
	RF	**97.03**	96.00	**1.96**	**97.96**	96.15
	GBM	**97.03**	**97.00**	2.94	97.00	**97.06**
	LDA	57.92	71.00	54.90	55.91	61.33

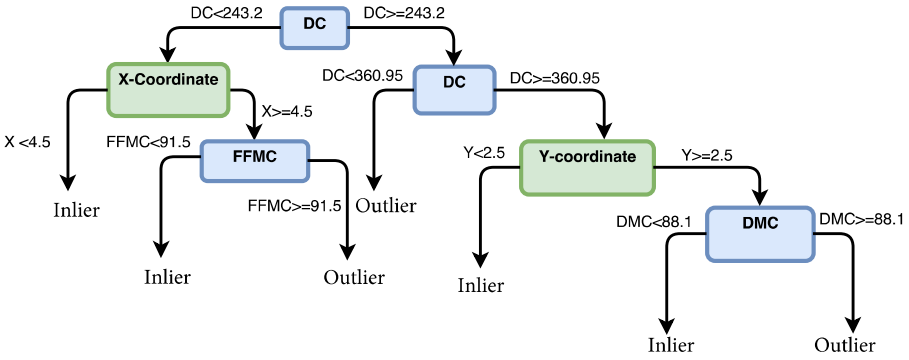


Fig. 4: Outlier detection CART model for STFWI in Event detection

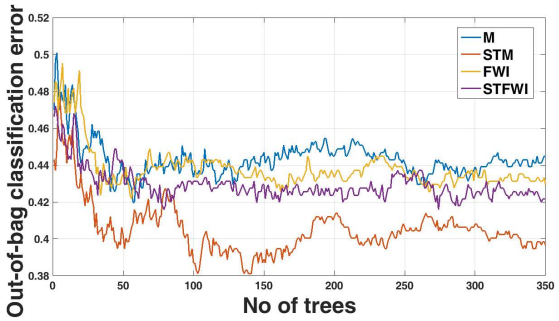


Fig. 5: Out-of-bag classification error for Event detection using RF

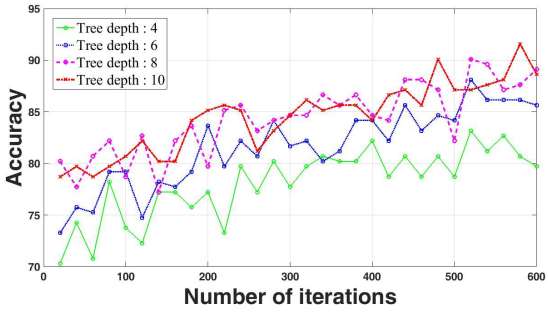


Fig. 6: Performance accuracy of GBM model for STM against the number of boosting iterations for various tree depths in Event detection

is further observed that for one instance RF has 100% accuracy, but overall GBM gives more promising results. LDA's performance is comparatively lower that the other models. One reason of this could be that LDA's assumption of the underlying distribution of data to be normal is not foolproof in this regard as any presence of outlier in the dataset contradicts this assumption and therefore greatly reduces the model's performance.

C. Analysis of Event detection

To achieve our goal of identifying rare events, the environmental parameters are explored. These parameters extracted from the dataset determine the set of attributes given as inputs to the learning models. We have compared the performance of the four classification models, by examining the impact of four distinct feature selection setups related with the occurrence of fire; mentioned in the first column of Table I. Table II illustrates the performance for Event detection

where apart from accuracy several other performance metrics such as Recall, FPR, Precision and Specificity were also used. It is observed that all the models produce remarkable results except for LDA. The best performance is marked by bold-face against each feature selection. For Event detection, RF could be considered the best model which generates zero False Positive results for STFWI and STM with Precision of 100% in both cases. This provides our evidence to the fact that spatio-temporal dependencies are to be considered for efficient detection of rare events as sensor data are known to exhibit strong spatial and temporal correlation. That is, readings generated from close distant neighbours tend to be similar and readings at a specific time instant are close to the past readings of the same phenomenon. Performance of CART is also satisfactory with upto 87.76% accuracy (M). Fig. 4 shows the CART model for a particular case STFWI feature selection. The root node in the figure is given by DC which also indicates its most significance status among all the other features. Prediction of the outliers is carried out by first testing each node that indicates the direction of the next branch to explore, until the leaf node is reached which generates the predicted outcome of whether an input is an outlier or not. The out-of-bag classification errors for RF is depicted in Fig. 5 which shows that error seem to remain constant after 300 trees. Fig. 6 shows the performance of GBM model by varying the tree depths where it is observed that the best accuracy corresponds to tree-depth = 10. For this reason, we have considered tree-depth=10 for experimentation. Also, since no significant increase in accuracy was observed after number of boosting iterations = 350, we have wisely selected this value in our experiment.

V. Conclusion

Since IoT based monitoring applications are becoming increasingly popular today, it becomes indispensable to ensure that the data gathered be accurate, reliable and uncorrupted before any critical decision about the monitoring environment is taken. Since outliers in the collected data greatly degrade its quality, in this work we have adopted four statistical learning approaches CART, RF, GBM and LDA for the correct identification of outliers. IoT systems are primarily composed of sensors that generates raw data from the physical environment and are also capable of detecting critical events in the sensing environment. However, it is often difficult to fathom the cause of the occurrence of outliers. As, outliers could be an outcome of either a malfunctioning sensor or an important event. Thus in our work, we have separately experimented on both Error and Event with the four models taking into account both spatial and temporal characteristics. For the purpose of Error detection, univariate parameters are taken to detect faulty sensor nodes. In case of Event, four distinct feature selection is done for conducting experiments. Simulation results show that the solution can effectively detect both types of outliers with upto 100% accuracy for Error detection (wind using RF) and upto 98.51% in case of Event detection (STM using RF).

VI. Acknowledgement

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